

USEFUL FORMULAE & CONCEPT

ALGEBRA

CAT QUANTITATIVE APTITUDE

$$y = mx + b$$

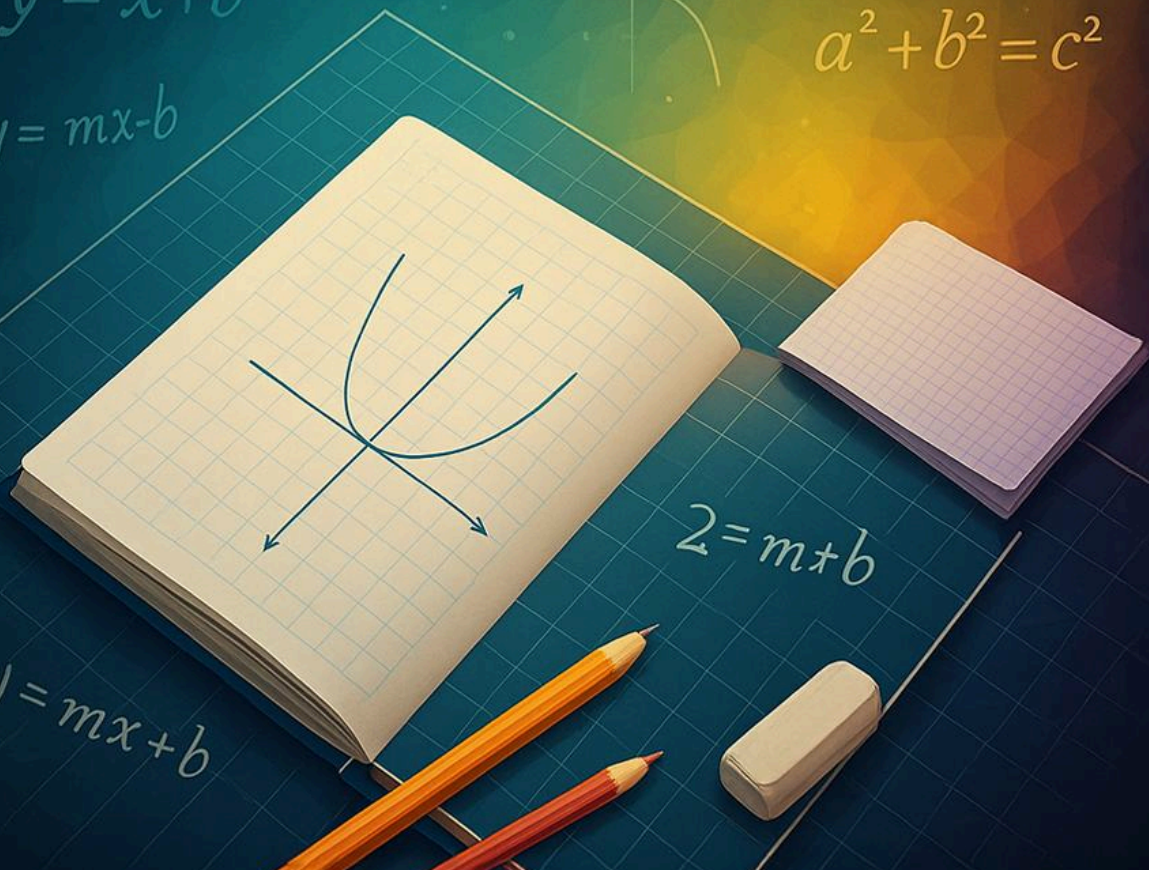
$$a^2 + b^2 = c^2$$

$$y = x + b$$

$$y = mx - b$$

$$z = m + b$$

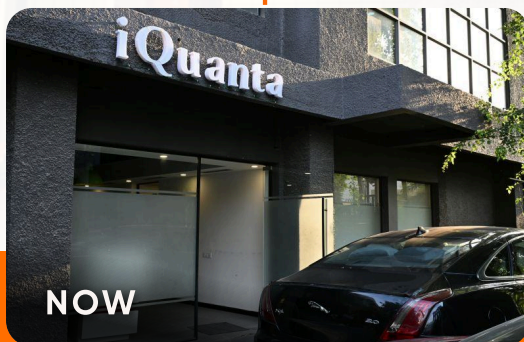
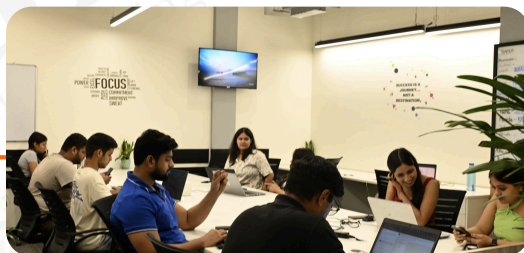
$$y = mx + b$$



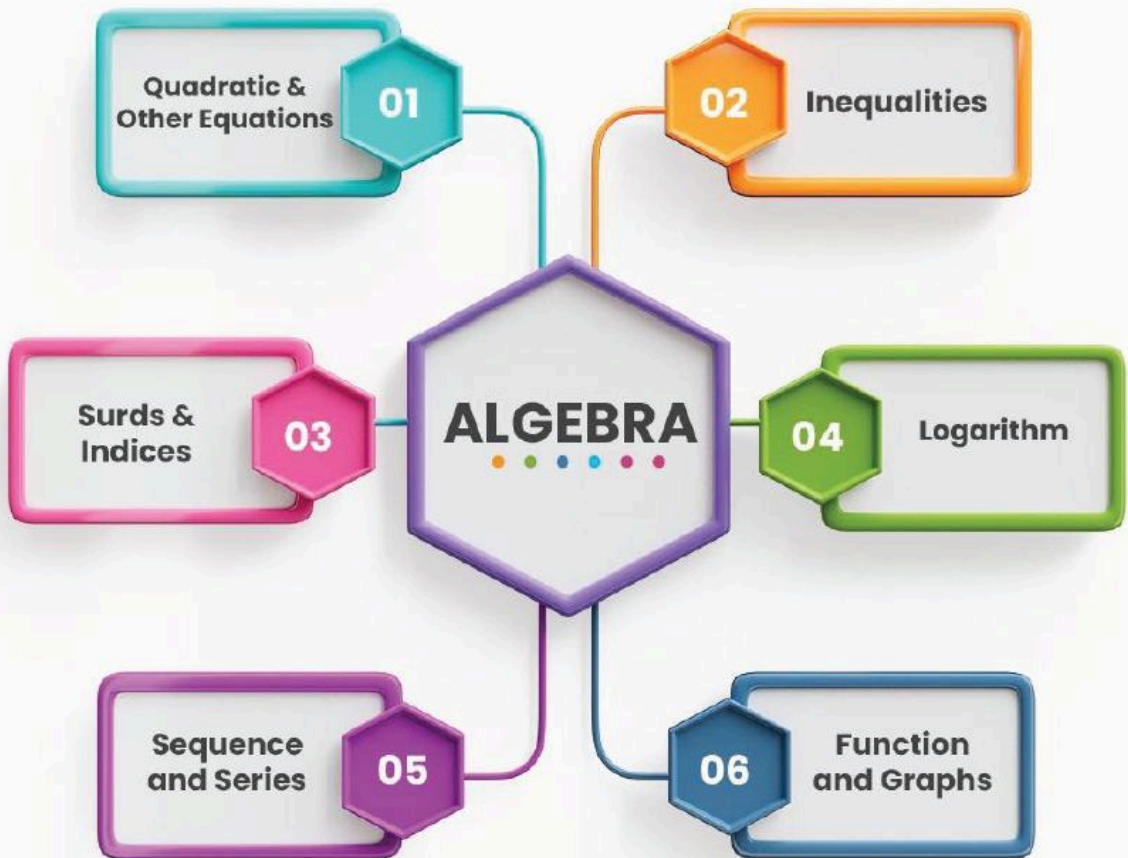
About iQuanta?

Founded in 2015 by Indrajeet Singh, iQuanta began as a Facebook doubt-solving group from a small apartment. With strong expertise in Quants, he earned the title 'Wizard of Quant' by solving thousands of students' queries with clarity and speed. Noticing the lack of quality guidance for CAT aspirants, he officially launched iQuanta to bridge this gap.

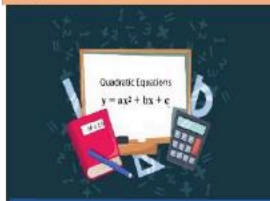
Over the years, iQuanta has continued to support over 4 lakh students through its active online communities, offering resources, and peer-driven learning. In the last 5 years, more than 8,000 students have converted IIMs. In 2024, iQuanta produced 500+ 99 percentilers, many of whom made it to top IIMs. iQuanta continues to guide thousands of aspirants on their journey to top B-schools.



Indrajeet Singh
Founder & CEO, iQuanta
Forbes 30 Under 30



QUADRATIC & OTHER EQUATIONS



Some important Algebraic Identities:

- $(a + b)^2 = a^2 + b^2 + 2ab$
 - $(a - b)^2 = a^2 + b^2 - 2ab$
 - $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
 - $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
 - $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
 - $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
 - $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
 - $a^2 - b^2 = (a + b)(a - b)$
 - $(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(c + a) + 3abc$
 - $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
- If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$

A quadratic equation is represented as

$$\boxed{ax^2 + bx + c = 0} \quad (\text{if } a = 0, \text{ then equation becomes linear})$$

$$\boxed{X = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$



All values of 'x' satisfying the equation are known as roots/zeros of the equation.

Note: An equation of degree '**n**' will have n roots (real and imaginary) (E.g. Cubic equation has 3 roots)

- If 'c' and 'a' are equal then the roots are reciprocal to each other.
- A quadratic whose roots are reciprocal of the roots of $ax^2+bx+c = 0$ is $cx^2+bx+a = 0$
- If $b = 0$, then the roots are equal and are opposite in sign
- **Discriminant :**

It is denoted by **D**, and $\mathbf{D = b^2 - 4ac}$. Depending on the sign and value of D, nature of the roots would be as follows:

- If $\mathbf{D < 0}$, Roots will be imaginary. The graph would not touch x axis.
- If $\mathbf{D > 0}$, Roots will be real and distinct. Graph cuts X axis at two distinct points.
- If $\mathbf{D = 0}$, Roots are real and equal. Graph just touches the x axis at one point.

Note: Complex or irrational roots always occur in pair i.e., they are conjugate. E.g.: $x^2 - 4x - 1 = 0$

$x = (2 \pm \sqrt{5})$ i.e., roots are $2 + \sqrt{5}$ and $2 - \sqrt{5}$ which are conjugate pairs.

	Quadratic Equation $\mathbf{ax^2 + bx + c = 0}$	Cubic Equation $\mathbf{ax^3 + bx^2 + cx + d = 0}$
Sum of roots	$\alpha + \beta = -b/a$	$\alpha + \beta + \gamma = -b/a$
Product of roots	$\alpha\beta = c/a$	$\alpha\beta\gamma = -d/a$
Pairwise sum of Product of roots	-	$\alpha\beta + \beta\gamma + \lambda\alpha = c/a$
A quadratic eq. can be written as $\mathbf{x^2 - Sx + P = 0}$ where S = sum of roots and P = product of roots		

- For any given equation $y = f(x) = 0$ the number of times the graph of this equation cuts the X axis is equal to the distinct real roots of this equation. For Exp: $(x-1)(x+2)(x-2) = 0$ will intersect x axis at 3 distinct points: 1, -2, 2
- Any quadratic equation will be of the form $(x-a)(x-b) = 0$ and will cut the axis at a and b.
- When $a > 0$, $ax^2 + bx + c$ has minimum at $x = -b/2a$ & that minimum is $-D/4a = (4ac-b^2)/4a$
- When $a < 0$, $ax^2 + bx + c$ has maximum at $x = -b/2a$ & that maximum is $-D/4a = (4ac-b^2)/4a$

Initially confused, impressed by **iQuanta's Doubt Solving**, gets **CAT 99.03%ile** with **iQuanta**



Karal Maheshwari

4 Jan 2020 · 🌐

99.03%ile, Jitna expected tha usse kam aaya but want to thank Indrajeet Singh sir 🙏🙏
Left my job in june and started preparing for cat, was confused about joining online coaching, but then I saw sir clearing doubts of students in iQuanta group and after 10 days i decided to join iQuanta, it was a big step for me cause I had to give my best. Thanks to iQuanta team and especially indrajeet sir for great sessions and proper direction one needs for exam like CAT ❤️ I am not in a position to take a drop else I would have with iQuanta team besides me.

700+ 99%ilers & 1000s of CAT Toppers



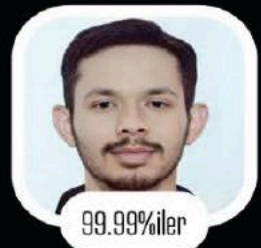
Rishi Mittal



Sabyasachi



Mridul Agarwal



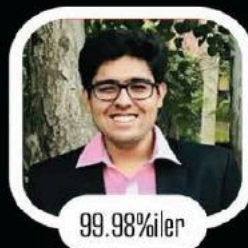
Parikshit Tomar



Guru Charan



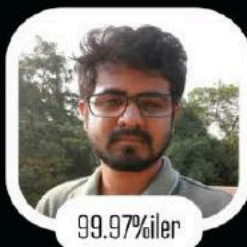
Manav Jain



Shikhar Sachdeva



Mihir Kapse



Satyajit Das



Swadesh Rath



Kislay Jha



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INEQUALITIES

If Roots are Real,

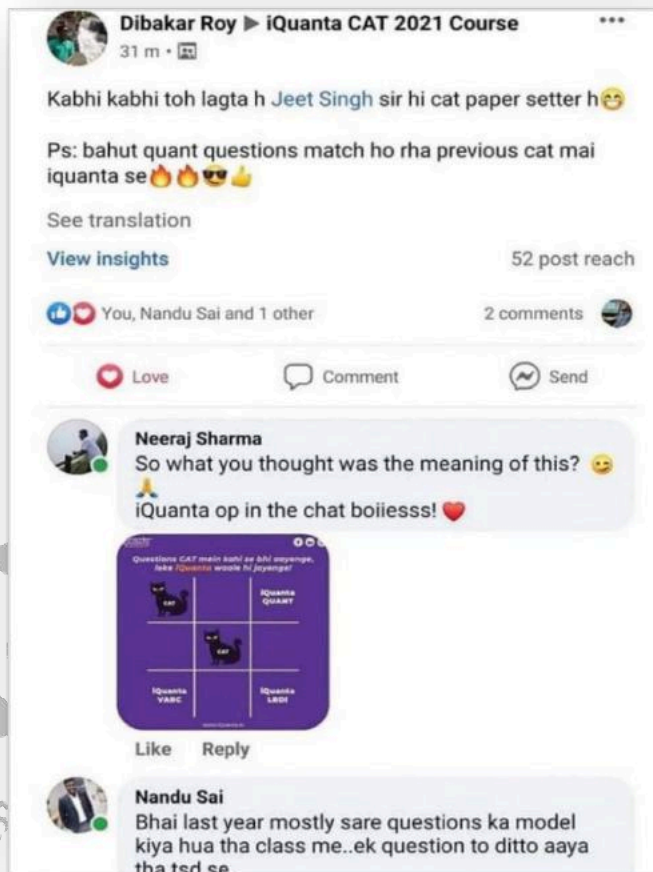
- when $(x-a)(x-b) > 0$
 $x \in (-\infty, a) \cup (b, \infty)$
- when $(x-a)(x-b) < 0$
 $x \in (a, b)$

Note:

- If product of n positive numbers is constant, then their sum will be minimum when all are equal or close to each other.
- If the sum of n positive numbers is constant, then their product will be maximum when all are equal or close to each other.
- In case of equations, for example $3x = 5y$, multiplying with -1 on both sides results in $-3x = -5y$.

But in the case of inequalities, for example $3x > 5y$, on multiplying with -1 on both sides, the inequality changes to $-3x < -5y$
For example $7 > 4$, but $-7 < -4$.

- In inequalities, one cannot cancel the common multiple on both sides. For example, cancelling the common multiple 'x' in $x(x-1) > x(y-2)$ is not allowed.



SURDS & INDICES

Important points to remember:

- In surd questions, every surd is an irrational number.
Ex: $\sqrt{4}$, $\sqrt[2]{5^3\sqrt{7}}$
- Any integer raised to the power zero will always equal one.

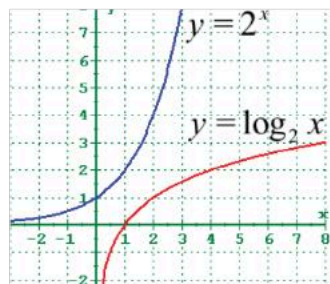
Law of surds	Law of indices
$\sqrt{m \times n} = \sqrt{m} \times \sqrt{n}$	$a^n \cdot a^m = a^{m+n}$
	$a^n \cdot b^n = (a \cdot b)^n$
$\sqrt{\frac{m}{n}} = \frac{\sqrt{m}}{\sqrt{n}}$	$a^m / a^n = a^{m-n}$
	$a^n / b^n = (a / b)^n$
$\frac{r}{\sqrt{s}} = \frac{r}{\sqrt{s}} \times \frac{\sqrt{s}}{\sqrt{s}} = \frac{r\sqrt{s}}{s}$	$(a^n)^m = a^{n \cdot m}$
$p\sqrt{q} \pm r\sqrt{q} = \sqrt{q}(p \pm r)$	$a n^m = a^{(n^m)}$
	$m\sqrt{(a^n)} = a^{n/m}$
$\frac{r}{p \pm q\sqrt{n}}$: Multiply num and den by $p \mp q\sqrt{n}$ to rationalise the denominator	$n\sqrt{a} = a^{1/n}$
	$a^{-n} = 1 / a^n$

LOGARITHMS

- $a^x = N$ can be expressed in logarithmic form as $x = \log_a N$
 $\log_a a = x$ means that $a = 10^x$
- Natural Logarithm:** $\log_e N$ is called Natural logarithm, denoted by $\ln N$ i.e., when the base is 'e' then it is called as Natural logarithm.
- Common Logarithm:** $\log_{10} N$ is called **common logarithm** i.e., when base of log is 10, then it is called as common logarithm.
- Both functions are graphed below: Base of 2 and base of e**

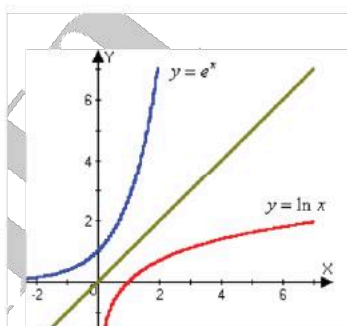
Logarithm Properties:

- $\log(mn) = \log m + \log n$, $m > 0$, $n > 0$ **(Addition)**
- $\log\left(\frac{m}{n}\right) = \log m - \log n$, $m > 0$, $n > 0$ **(Subtraction)**
- $\log m^n = n(\log m)$ **(Logarithm of a power)**
- $\log_x y = \frac{\log_a y}{\log_a x}$ **(Change of base rule)**
- $\log_x y = \frac{1}{\log_y x}$ **(Inverse)**
- $\log_x 1 = 0$ ($x \neq 0, 1$).



Note: The logarithm of "0" and negative numbers is not defined.

-
- $\log_b 1 = 0$ ($\because b^0 = 1$)
- $\log_b b = 1$ ($\because b^1 = b$)
- $y = \ln x \rightarrow x = e^y$
- $x = e^y \rightarrow \ln x = y$
- $x = \ln e^x = e^{\ln x}$
- **$\log_b b^y = y$**



SEQUENCES, SERIES & PROGRESSIONS



Arithmetic Progression (A.P)

An A.P. is of the form $a, a+d, a+2d, a+3d, \dots$ where a is called the 'first term' and d is called the 'common difference'.

n^{th} term of an A.P:

$$T_n = n^{\text{th}} \text{ term} = a + (n - 1) d$$

(d = common difference, n = numbers of terms)

Sum of the first n term of an A. P:

$$S_n = \frac{n}{2} [2a + (n - 1)d] = \frac{n}{2} [a + L]$$

(L = last term, a = first term)

iQuanta CAT 2024 IIM Converts

IIM Ahmedabad Converts



Vedant Chandewar
99.98%iler



Swaraj Pal
99.95%iler

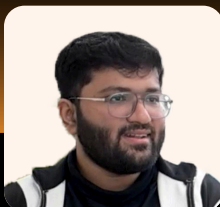


Raunak Das
99.55%iler



Siddhant Kumar
99.42%iler

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Mridul Tiwari
99.97%iler



Shaivi Goyal
99.54%iler



Rahul Sinha
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Aman Ehtesham
99.94%iler

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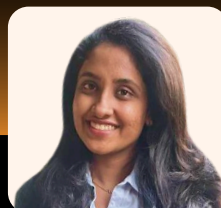
Mridul Tiwari
99.98%iler



Sanvie Singhal
99.92%iler



Chitturi Sakshvith
Sanjeev
99.86%iler



Madhumitha
Sriharsha
99.6%iler

And many more

Arithmetic Mean (A.M): $A.M = \frac{1}{2}(\text{Sum of all terms}) = \frac{1}{2}(\text{first term} + \text{last term})$

Geometric Progression (G.P)

A G.P. is of the form $a, ar, ar^2, ar^3, \dots$ where a is called the 'first term' and r is called the 'common ratio'.

n th term of a G.P:

$$T_n = a r^{n-1}$$

Sum of the first n terms in a G.P:

$$S_n = \frac{a(1-r^n)}{1-r}$$

; Sum of ∞ terms:

$$S_\infty = \frac{a}{1-r}$$

Geometric Mean (G.M):

$$b = \sqrt{ac}$$

(a, b, c are the 3 consecutive terms of a GP)

Harmonic Progression (H.P.)

If the reciprocals of the terms of a sequence are in arithmetic progression, the sequence is called harmonic progression.

For E.g. $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ is in harmonic progression. If we write it in form of H.P

i.e., $\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots$

Harmonic Mean (H.M) of $x, y = \frac{2xy}{x+y}$

For any two positive number a and b , $A.M \geq G.M \geq H.M$

Properties of H.P:

If $a, b, c, d, \dots$ are in H.P, then $a+b > b+c$ and $ad > bc$

Arithmetic Geometric Series:

- A series will be in arithmetic geometric series if each of its terms is formed by the product of the corresponding terms of an A.P and G.P.

- The general form of A.G.P series is $a, (a + d)r, (a + 2d)r^2, \dots$
- Sum of 'n' terms of A.G.P series

$$S_n = \frac{a - [a + (n-1)d]r^n}{1-r} + \frac{dr(1 - r^{n-1})}{(1-r)^2} \quad (r \neq 1)$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

- Sum of infinite terms of A.G.P series

$$S_n = \frac{a}{1-r} + \frac{dr}{(1-r)^2} \quad (|r| < 1)$$

- Relationship between A.M, G.M and H.M for two numbers a and b,

$$G.M = \sqrt{(A.M \times H.M)} \quad | \quad A.M \geq G.M \geq H.M$$

Some useful results:

- Sum of first n natural numbers = $\frac{n(n+1)}{2}$
- Sum of the squares of first n natural numbers = $\frac{n(n+1)(2n+1)}{6}$
- Sum of the cubes of first n natural nos. = $\left[\frac{n(n+1)}{2}\right]^2$
- Sum of first n natural odd numbers = n^2
- Sum of first n even numbers = $n(n+1)$

FUNCTIONS & GRAPHS

A function is a special relationship where each input has a single output (but 2 different inputs can have the same output). It is often written as " $y = f(x)$ " where x is the input value in the function y .

Domain: Set of real and finite values that x can take.

Range: Set of real and finite values that y can have corresponding to the values of x .

From **85%ile to 99.9%ile in CAT**, that's what relevant Content at iQuanta helps you to get :



Geetsaisumant Jupudi

21 December at 23:00 · 🌐

A journey from 85 percentile to 99.89 percentile.

My first CAT attempt in 2020 during my final year and landed up getting 85 percentile without any preparation.

I decided to appear for CAT again, but this time with preparation. I started to search for an online coaching institute and got to know about Iquanta through quora. I decided to join iQuanta(Second Batch) for two reasons, one was the positive reviews and second that it was pocket friendly.

I joined iQuanta and prepared for 5-6 months, but during last 2 months of CAT preparation I took frequent breaks, and finally got to 96.68 percentile. On the day response sheet was released, I realized that I had underperformed. I was still hopeful that I would get an admission in a decent college. But fortunately I didn't. Then somewhere around June 2022, I decided to give CAT a last attempt and give everything to it. I joined Iquanta course(Last Batch) again, this time because I was confident of the content and rigorous practice at Iquanta. I did not repeat the same mistake again and here it is, the day I was waiting for. I cannot explain the my feeling when I saw my result. 2 years long wait came to an end, still I believe far to go(GDPI), but one stage is cleared and after seeing the result, I am more pumped up for GDPI Preparation.

Most important thing that I learnt in during my journey is that CONSISTENCY is the key to crack CAT.

You will have bad days during preparation, I had, everyone has. But never get disheartened, believe in the process and work hard till the end.

The best part I feel about Iquanta is the huge amount of RELEVANT content, whether it is LRDI sets, RCs or Quant questions, everything is at par with CAT level. Then the rigorous practice and peer learning complement your preparation and keep you motivated.

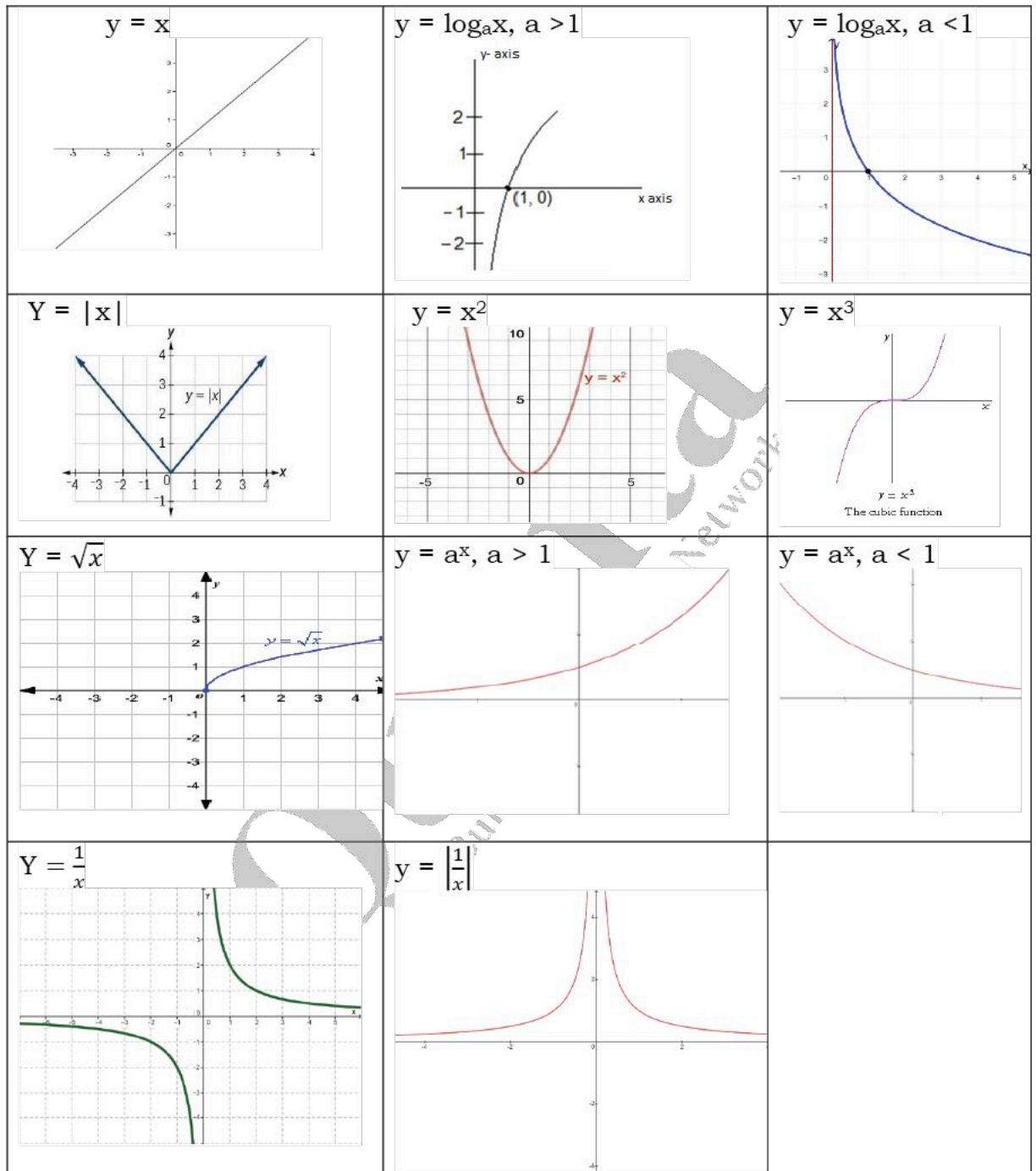
NOTE- Those who are planning to take Iquanta course, don't miss the final marathon sessions, those are score boosters.

Saying thanks won't be enough to express my gratitude to Jeet Singh sir, Sajjan Barnwal sir, Abhishek Leela Pandey Sir and all the iQuanta team, who guided us throughout our journey and kept us motivated. Couldn't have got a better institute that this.



Note: Set A has m elements & set B has n elements then no. of functions possible from set A to set B = n^m

Some important graphs:



iQuanta vs OTHER COACHINGS

30,000 IIM Calls from iQuanta
CAT Course

COURSE FEATURES	CAT FULL COURSE	CAT FULL COURSE PRO	OTHER COACHINGS
Live Conceptual Classes	✓	✓	✓
Application Classes	✓	✓	✗
All Year Practice Sessions	✓	✓	✗
Assignments	✓	✓	✓
24*7 Doubt Solving	✓	✓	✗
Full Mocks with Video Solution	✓	✓	✓
Sectionals with Video Solution *	✓	✓	✗
Non-Engineer's Quant *	✓	✓	✗
Engineer's VARC *	✓	✓	✗
LRDI Inception *	✓	✓	✗
AI Mock Analysis	✓	✓	✗
Special Initiatives * (QA250, LRDI 70)	✓	✓	✗
Peer to Peer Learning	✓	✓	✗
Free 2 Month Crash Course	✓	✓	✗
7500 CAT Qs in IIM ABC Course	✗	✓	✗
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12 Sets: CAT Level Books *	✗	✓	✗

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